

# Tiling For Random Matrices With Asymmetric Precision

Project Aleph Final Report

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## Contents

|          |                                            |          |
|----------|--------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                        | <b>2</b> |
| 1.1      | Tiled Matrix Multiplication . . . . .      | 2        |
| 1.2      | Reasons To Study Random Matrices . . . . . | 2        |
| <b>2</b> | <b>Simulator</b>                           | <b>2</b> |
| 2.1      | Architecture . . . . .                     | 2        |
| 2.2      | Methodology . . . . .                      | 2        |
| <b>3</b> | <b>Optimal Tiling</b>                      | <b>2</b> |
| 3.1      | Traffic-Optimal Tiling . . . . .           | 2        |
| 3.2      | PRNG-FIFO . . . . .                        | 2        |
| 3.2.1    | Choosing a Dataflow . . . . .              | 2        |
| 3.2.2    | Generating B vs Loading B . . . . .        | 2        |
| 3.3      | Traffic-Optimal vs Cycle-Optimal . . . . . | 3        |
| 3.4      | Calibrating $\alpha$ . . . . .             | 3        |
| 3.4.1    | Is This Correct? Yes . . . . .             | 3        |
| 3.4.2    | Optimal Tile Shape vs $g\_c$ . . . . .     | 3        |
| <b>4</b> | <b>Impact of Tile Selection</b>            | <b>3</b> |
| 4.1      | Optimal vs Square Tiles . . . . .          | 3        |
| <b>5</b> | <b>Conclusion</b>                          | <b>3</b> |

# 1 Introduction

why asymmetric precision and a randomly generated B change the tiling problem, in both memory traffic and cycle count

## 1.1 Tiled Matrix Multiplication

C-stationary tiling: keep a C tile in L1 and stream an A row-band and a B col-band through it

## 1.2 Reasons To Study Random Matrices

why random matmul with asymmetric precision is interesting, bring examples of randomized algos

# 2 Simulator

## 2.1 Architecture

## 2.2 Methodology

hardware config, workload sizes, modeling assumptions, matrix registers, matmul,  $T_k = A\_width = B\_height$ . also L1 should be tuned to be fully associative and we should be able to see exactly where on graphs L1 overflow happens

# 3 Optimal Tiling

## 3.1 Traffic-Optimal Tiling

validate the traffic-optimal asymmetric tile ( $1/\rho$ )

- experiment: `v5-results/paper-model/paper-traffic-model`
- experiment: `v5-results/paper-model/paper-per-matrix-balance`

## 3.2 PRNG-FIFO

explain how prng-fifo works, the fact that it works in parallel with memory accesses, mention negligible seed traffic - experiment: not sure there is experiment, but need to sweep over  $g\_c$  for b-stationary and c-stationary modes and check cycle count. **important!  $g\_c$  needs to be measured not just in cycles, but has to be compared to some other cycle cost in the simulation.** for example: plot how much cycles it costs to bring an element of A vs how much it costs to bring an element of B (i.e. sweeping over  $g\_c / T\_m$ )

### 3.2.1 Choosing a Dataflow

b-stationary vs c-stationary, fifo streams B in fixed order, b-stationary reuses each B element  $T\_M$  times

- experiment: `v55-results/b-stationry-vs-c-stationary`
- experiment: `v55-results/b-stationary-vs-c-stationry-col-major` (maybe?? i dont get totally what this does)

### 3.2.2 Generating B vs Loading B

when does FIFO generation beat bringing B from memory

- experiment: `v5-results/math-model-no-l2/e13-fifo-vs-mem`

### 3.3 Traffic-Optimal vs Cycle-Optimal

minimizing traffic != minimizing cycles, why the old model doesn't work, new formula. put graphs that show that it's true

- experiment: `v5-results/prng-fifo-pipelined-sweep`
- experiment: `v5-results/math-model-no-l2/e14-pipelined-fifo-vs-mem`

### 3.4 Calibrating $\alpha$

- experiment: `v5-results/math-model-no-l2/e6-tn-independence`. **important!** have to tie Tile-M dimension to some size in the simulation, probably to the size of L1
- experiment: `experiments/v5-results/math-model-no-l2/e3-alpha-calibration`. same notion

#### 3.4.1 Is This Correct? Yes

if we calibrate  $\alpha$  once for  $g\_c=0$ , we will find that it's optimal for  $g\_c > 0$

- experiment: `v5-results/math-model-no-l2/e8-gc-boundary-sweep`

#### 3.4.2 Optimal Tile Shape vs $g\_c$

measured best tile per  $g\_c$  (not model prediction). maybe need a longer x-axis sweep? need to understand the results

- experiment: `v5-results/math-model-no-l2/e8-gc-boundary-sweep` (`best_shape_per_gc.png`)

## 4 Impact of Tile Selection

### 4.1 Optimal vs Square Tiles

we have a speedup. the y-axis needs to be better formulated

- experiment: `v5-results/math-model-no-l2/e8-gc-boundary-sweep` (`optimal_vs_square.png`)
- experiment: `v5-results/math-model-no-l2/e8-gc-boundary-sweep` + `e13-fifo-vs-mem` (`fifo_vs_mem_square.png`)

## 5 Conclusion

asymmetric tile beats square in both traffic and cycles, one-shot  $\alpha$  calibration predicts the best tile